

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

C03: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

**Time Duration: 1 Hour
Total Marks:30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

*I V.SURYA SATVIKA(192524254)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: V.S.SATVIKA

Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause ($\square$) if the goal is proved.

Absolutely. I've gone through the **CO3_AT3.pdf** and will answer all 5 questions exactly according to the instructions, using the **Resolution Algorithm**, including propositional representation, CNF, negated goal, resolution steps, empty clause, and conclusion. The assessment is for **Artificial Intelligence (CSA17), CO3**, and specifically requires resolution-based logical reasoning.

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that the ground is wet using the Resolution Algorithm.

1. Propositional Logic Representation

Let:

- **R** = It is raining.
- **W** = Ground is wet.

The statements are:

1. If it rains, the ground becomes wet:

$$R \rightarrow W$$

2. It is raining:

R

Goal:

W

2. Convert to CNF

The implication:

$$R \rightarrow W$$

is equivalent to:

$$\neg R \vee W$$

Therefore, the CNF clauses are:

- **C1: $\neg R \vee W$**
- **C2: R**

3. Negate the Goal

$$\text{Goal} = W$$

Negated goal:

$$\neg W$$

So we have:

- **C1: $\neg R \vee W$**
- **C2: R**
- **C3: $\neg W$**

4. Resolution Steps

Resolve C1 and C2:

$$\begin{array}{l} \neg R \vee W \\ R \end{array}$$

Therefore:

$$W$$

New clause:

$$\mathbf{C4: W}$$

Now resolve C4 with C3:

$$\begin{array}{l} W \\ \neg W \end{array}$$

Therefore:

□

5. Empty Clause

$$W + \neg W \rightarrow \square$$

Conclusion

Since the **Empty Clause** ($\square$) is obtained, the negation of the goal leads to a contradiction.

Therefore:

The ground is wet.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, the student receives internal marks.
2. Rahul submitted the assignment.

Goal

Prove that Rahul receives internal marks.

1. Propositional Logic Representation

Let:

- **S** = Rahul submitted the assignment.
- **M** = Rahul receives internal marks.

Statements:

1. $S \rightarrow M$

2. S

Goal:

M

2. CNF Conversion

Convert:

$S \rightarrow M$

to:

$\neg S \vee M$

Therefore:

- $C1: \neg S \vee M$
- $C2: S$

3. Negate the Goal

Goal:

M

Negated goal:

$\neg M$

Clauses:

- $C1: \neg S \vee M$
- $C2: S$
- $C3: \neg M$

4. Resolution

Resolve C1 and C2:

$\neg S \vee M$

S

Therefore:

M

New clause:

C4: M

Now resolve C4 and C3:

M

$\neg M$

Therefore:

□

5. Conclusion

The Empty Clause is derived:

$M + \neg M \rightarrow \square$

Hence, the negated goal is contradictory.

Therefore, Rahul receives internal marks.

Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member. **Goal**

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, the person can borrow books.
2. Priya is a library member.

Goal

Prove that Priya can borrow books.

1. Propositional Logic Representation

Let:

- **L** = Priya is a library member.
- **B** = Priya can borrow books.

Statements:

1. **$L \rightarrow B$**
2. **L**

Goal:

B

2. CNF Conversion

Convert:

$L \rightarrow B$

to:

$\neg L \vee B$

CNF clauses:

- **C1: $\neg L \vee B$**
- **C2: L**

3. Negate the Goal

Goal:

B

Negated goal:

$\neg B$

Therefore:

- **C1: $\neg L \vee B$**
- **C2: L**
- **C3: $\neg B$**

4. Resolution Steps

Resolve C1 and C2:

$\neg L \vee B$
 L

Therefore:

B

New clause:

C4: B

Resolve C4 and C3:

B
 $\neg B$

Therefore:

□

5. Final Conclusion

Since the Empty Clause is obtained:

B + $\neg B \rightarrow \square$

Therefore:

Priya can borrow books.

The goal is successfully proved using the Resolution Algorithm.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

Prove that Arun is eligible for placement.

1. Propositional Logic

Let:

- **A** = Arun cleared the aptitude test.

- **P** = Arun is eligible for placement.

Statements:

1. **A** $\rightarrow$ **P**
2. **A**

Goal:

P

2. CNF Clauses

Convert the implication:

A $\rightarrow$ **P**

Therefore:

$\neg A \vee P$

CNF clauses:

- **C1: $\neg A \vee P$**
- **C2: A**

3. Negate the Goal

Goal:

P

Negated goal:

$\neg P$

Therefore:

- **C1: $\neg A \vee P$**
- **C2: A**
- **C3: $\neg P$**

4. Resolution Algorithm

First resolution:

C1: $\neg A \vee P$
C2: A

Resolving $\neg A$ and A gives:

C4: P

Second resolution:

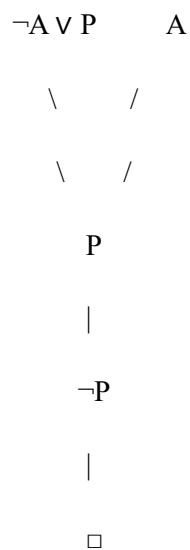
C4: P

C3: $\neg P$

Therefore:

□

5. Resolution Tree



Conclusion

The resolution process produces the **Empty Clause** ($\square$).

Therefore:

Arun is eligible for placement.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password. **Goal**

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause ($\square$).

5. Explain each resolution step and write the final conclusion.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, the user is authenticated.
2. If a user is authenticated, the user is granted access.
3. The user entered the correct password.

Goal

Prove that the user is granted access using the Resolution Algorithm.

1. Propositional Logic Representation

Let:

- **P** = User entered the correct password.
- **A** = User is authenticated.
- **G** = User is granted access.

Statements:

1. $P \rightarrow A$
2. $A \rightarrow G$
3. **P**

Goal:

G

2. Convert Implications into CNF

For statement 1:

$P \rightarrow A$

becomes:

$\neg P \vee A$

For statement 2:

$A \rightarrow G$

becomes:

$\neg A \vee G$

Therefore:

- **C1: $\neg P \vee A$**
- **C2: $\neg A \vee G$**
- **C3: P**

3. Negate the Goal

Goal:

G

Negated goal:

$\neg G$

So the complete set of clauses is:

- **C1: $\neg P \vee A$**
- **C2: $\neg A \vee G$**
- **C3: P**
- **C4: $\neg G$**

4. Resolution Steps

Step 1

Resolve C1 and C3:

$\neg P \vee A$
P

Therefore:

A

New clause:

C5: A

Step 2

Resolve C2 and C5:

$\neg A \vee G$
 A

Therefore:

G

New clause:

C6: G

Step 3

Resolve C6 and C4:

G
 $\neg G$

Therefore:

□

5. Resolution Tree

$$\begin{array}{c} \neg P \vee A \quad P \\ \backslash \quad / \\ \backslash \quad / \\ A \\ | \\ \neg A \vee G \\ | \\ G \\ | \\ \neg G \\ | \\ \square \end{array}$$

6. Final Conclusion

The Resolution Algorithm derives:

G

and then resolves it with the negated goal:

$$G + \neg G \rightarrow \square$$

Since the **Empty Clause** ($\square$) is obtained, the negated goal is false.

Therefore:

The user is granted access.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation